

STATEMENT OF MASTER'S THESIS ASSESSMENT

Master's Programme in Computer, Communication and Information Sciences

Author of the thesis: Cuong Nguyen
Title of the thesis: KEMEDHOC*: A formally verified quantum-safe key exchange protocol
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Location: Huawei Technologies Oy

This thesis sits in the area of formal verification of cryptographic key exchange protocols, specifically a post-quantum-secure variant of the EDHOC protocol. EDHOC is a computationally lightweight key exchange protocol published in IETF RFC 9528 which is especially suitable for constrained devices. This thesis identifies and addresses two shortcomings of EDHOC: (1) It does not offer security against quantum adversaries, and (2) only its design but not implementation is formally verified. It addresses the two issues by (1) proposing a variant of the EDHOC design called KEMEDHOC which is compatible with standardized post-quantum security key encapsulation mechanisms (KEM), and (2) performing formal verification on both the KEMEDHOC design and an implementation.

Definition of research scope and goals: 5

The thesis defines its research scope and goals with clarity and precision. It identifies two well-founded research gaps in the current EDHOC protocol: (1) post-quantum migration and (2) end-to-end design-to-code formal verification. These gaps are motivated by the increasing relevance of quantum-resistant cryptography and the need for verifiable assurance in IoT key-exchange implementations.

The research goals are formulated as two complementary objectives: designing and verifying a post-quantum variant of EDHOC (KEMEDHOC) and constructing a verified implementation pipeline using F*, Low*, and KaRaMel. The design of the post-quantum variant is formally proved using ProVerif, ensuring cryptographic soundness at the protocol level.

The thesis recognizes the emerging need for post-quantum EDHOC as noted in IETF RFC 9528, which references KEM-based methods combined with digital signatures but provides no concrete proposal. This work advances beyond that direction by designing and verifying a pure KEM-based approach, which is more efficient and conceptually cleaner than hybrid KEM + signature constructions.

The scope is ambitious yet well bounded within the domain of lightweight key exchange for constrained environments. The author demonstrates a clear understanding of how the defined problems connect to real-world cryptographic and IoT security challenges.

Command of the topic: 5

The thesis provides a comprehensive and accurate explanation of symbolic and computational models, the Dolev-Yao attacker model, and formal verification tools such as ProVerif, F*, Low*, and KaRaMel. These concepts are not only described but also applied correctly within the thesis' verification pipeline.

The work shows authors deep familiarity with the relevant standards and literature, including the IETF LAKE working group drafts, post-quantum cryptography research, and prior verified implementations such as miTLS and HACLS*. The sources are authoritative and current, ranging from foundational works in cryptographic proofs to 2025 IETF and IACR papers, reflecting excellent coverage and integration.

The author's understanding is not limited on the abstract theory of protocol verification, it extends implementation-level details such as memory safety, side-channel mitigation, and verified compilation. The thesis is cohesive and logically structured, maintaining conceptual continuity between formal theory and engineering practice.

Methods: 5

The thesis applies a rigorous and well-structured methodological framework that integrates protocol design, formal verification, verified implementation, and empirical evaluation. The author's approach demonstrates a mature understanding of how to combine theoretical and practical research methods within secure protocol engineering.

The methods are appropriately chosen for the research questions. The author created a formal ProVerif model for the newly designed purely KEM-based EDHOC extension, enabling symbolic verification of authentication and secrecy properties. The use of F* and ProVerif ensures mathematically sound reasoning about protocol correctness and cryptographic security. The verified implementation in Low* and KaRaMel bridges these proofs to executable C code, maintaining end-to-end correctness guarantees.

The work is not limited to formal proofs: the author also conducted benchmarking experiments that demonstrate the practical feasibility of the proposed KEM-based extension. The performance results show that the proposed variant is faster than the hybrid KEM + Falcon signature-based EDHOC, confirming both theoretical soundness and implementation efficiency.

The methods are executed independently and transparently, with clear documentation of assumptions, tools, and verification boundaries. Limitations—such as partial component verification and limited device-level testing—are critically discussed, reflecting research maturity.

Results and contribution: 5

The thesis adequately addresses the defined research questions and objectives. The results are both technically and scientifically sound, demonstrating the author's ability to produce original, verifiable, and practically relevant contributions in the field of secure protocol design.

The main results include:

- Design and formal verification of KEMEDHOC, a post-quantum-ready variant of the EDHOC protocol that preserves lightweight-ness and compatibility with IETF standards. The formal ProVerif model proves authentication and secrecy for the new purely KEM-based construction.
- End-to-end verified implementation pipeline, connecting formal proofs in F* and ProVerif to executable C code through Low* and KaRaMel, ensuring correctness-by-construction.
- Integration and empirical validation within the μ EDHOC codebase, accompanied by performance benchmarks confirming that the proposed pure KEM-based EDHOC is faster than the hybrid KEM + Falcon signature variant.

The topic of post-quantum EDHOC has recently gained momentum in the IETF community, and this thesis provides timely and valuable input for ongoing standardization. The proposed pure KEM-based method is particularly advantageous for constrained IoT devices, as it eliminates the need to support both a

KEM and a digital signature algorithm, resulting in reduced computational footprint and simpler implementation. The approach is also conceptually cleaner, aligning well with emerging KEM-centric cryptographic paradigms.

The author's contribution is substantial and independent, combining formal security proof, verified implementation, and empirical benchmarking. The results are clearly presented, validated, and critically interpreted in relation to the research questions.

Overall, the thesis makes an original and significant contribution to the field, offering both theoretical advancement and practical impact that are directly relevant to contemporary IETF standardization and the future of post-quantum IoT security.

Presentation, language, and structure: 5

The thesis is structured in a logical way that is easy to follow. The introduction provides all necessary contexts, identifies the research gaps, overviews the methodology and results, and addresses limitations and related/concurrent works. Sections 2 and 3 provide a concise introduction to the background knowledge of EDHOC and formal verification. Sections 4-7 present the technical results in detail and Sections 8-9 conclude the thesis with relevant discussions.

Overall, the thesis is written in a language that is both precise and easy to understand. The author takes good care of readers who may not have expertise in some aspects of the thesis by providing helpful examples and remarks. There are minor typos and grammatical errors which do not affect understand.

Thesis process: 5

From my perspective as a supervisor, Cuong worked almost completely independently especially when it comes to the technical content. Occasional editorial comments are taken seriously and addressed promptly. The thesis was submitted slightly later than the expected time which is well within a reasonable range.

Overall assessment:

Overall, the thesis is of exceptional quality and deserves the maximum grade.

Grade: 5

3 Dec 2025



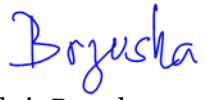
Russell W. F. Lai

Professor, thesis supervisor



Sampo Sovio

Thesis advisor

A handwritten signature in blue ink that reads "Brzuska". The signature is written in a cursive, flowing style.

Chris Brzuska

Professor, second reviewer